3,500 ft Rocket — Most Frugal Sourcing Guide

Cheapest verified source for every part across all three subsystems (flight computer, radio link, launch controller) plus airframe, recovery, and motor — built on top of the electronics viability research and the existing project sourcing reference.

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1. Flight Computer & Sensor Electronics

All viable per the electronics viability check. AliExpress is cheapest for nearly every part here because these are extremely commoditized, high-volume modules — the main cost is a 2–4 week wait, so order these first and in one consolidated cart to spread the shipping cost across more items.

Part	Cheapest Source	Frugal Price	Buying Notes
Microcontroller — Arduino Nano (clone, CH340)	AliExpress, multi-pack of 3–5	$2–4 each in a pack (~$10–15 for 5)	Buying a pack instead of one unit means your project-plan spare basically comes free. Single units on Amazon run $6–9 each.
Altitude — BMP388 barometric sensor	AliExpress (GY-BMP388 breakout)	$3–6	Amazon equivalent runs $8–12. Confirm the breakout includes onboard voltage regulation (nearly all do).
Accel/gyro — MPU6050	AliExpress	$1.50–3	The single cheapest part on the whole list — extremely commoditized. Amazon runs $5–8 for the same chip.
Position — NEO-6M GPS module	AliExpress	$6–10	Check the listing text for “unlocked” or absence of a COCOM altitude/velocity lock before buying. Amazon equivalents run $12–20.
Storage — MicroSD SPI module + card	AliExpress	$1–2 (module) + $4–6 (8–16 GB card)	Buy the 6-pin module with the onboard regulator + level-shifter chip visible in the photo, not the bare/raw socket — the raw version can damage the card on a 5-volt Nano.
Radio — 2× RFM95W LoRa module (915 MHz)	AliExpress	$4–7 each (~$8–14 for the pair)	Amazon runs $12–18 each for the same chipset. Double-check “915 MHz” is stated explicitly — a mismatched 433 MHz unit will not talk to its pair.
Power — buck/boost voltage regulator	AliExpress, multi-pack of 5	$1–2 for the pack	Cheap enough that spares cost nothing extra — buy the pack over a single unit.
Power — 1S LiPo battery (500–900 mAh)	Amazon or a domestic hobby/drone retailer	$6–10	Worth paying slightly more than AliExpress here: a battery is the one safety-relevant part on this list, and a domestic seller makes it easier to confirm a genuine built-in protection PCB and get a replacement quickly if a cell arrives swollen or damaged.

Subsystem subtotal (frugal, AliExpress-first): roughly $25–40, versus the $25–45 range in the original sourcing reference.

2. Launch Controller Components

This box is built once and reused on every future launch, so these are genuinely one-time costs. Generic electrical/hardware parts, not rocketry-specific — a local hardware store can beat online shipping on the bulkier items (switches, wire, enclosure), while the small electrical bits are cheapest in bulk online.

Part	Cheapest Source	Frugal Price	Buying Notes
Arm/safety key switch	Amazon or local hardware store	$4–8	A simple keyed toggle switch rated for at least a few amps is plenty — no need for a rocketry-branded version.
Momentary launch button	AliExpress or Amazon	$2–5	A weatherproof momentary push-button (the kind sold for doorbells or industrial panels) works well and is cheaper than a hobby-marketed “launch button.”
Continuity-check LED + resistor	AliExpress, multi-pack of 50+	Under $1 for the whole pack	Buy an assorted LED + resistor pack — you will use these across every future project, not just this one.
12 volt firing battery	Local hardware/battery store or Amazon	$10–20	A sealed lead-acid or a pack of 8 AA cells in a holder both work; buying locally avoids a shipping surcharge on batteries that many carriers apply.
Lead wire with alligator clips (20–30 ft / 6–9 m, 18–20 AWG)	Local hardware or automotive parts store	$8–15	Automotive-section jumper/test lead wire is usually cheaper per foot than anything marketed as “rocketry launch wire.”
Project enclosure box	Local hardware store or Amazon	$5–10	A basic plastic electrical junction box is fine and far cheaper than a purpose-built rocketry controller case.

Subsystem subtotal (frugal): roughly $30–60 one time, versus the $15–30 range in the original reference — the earlier estimate assumed a lighter-duty build; the range above assumes a sturdier reusable box meant to last for years of launches.

3. Airframe & Recovery Materials

Airframe parts are rocketry-specific and not available on general marketplaces at comparable quality, so the two hobby vendors already in the project plan remain the frugal choice — the savings here come from comparing prices between them and from a few DIY substitutions on recovery gear.

Part	Cheapest Source	Frugal Price	Buying Notes
Body tube — 66 millimeter (2.6 inch) outer diameter, phenolic	Apogee Rockets or Wildman Rocketry (compare both)	$15–25 per 3–4 ft (0.9–1.2 m) section	Wildman Rocketry is frequently a few dollars cheaper on the same phenolic stock — worth a direct price check before ordering, per the project's vendor-comparison note.
Nose cone — plastic ogive	Apogee Rockets or Wildman Rocketry	$8–15	Plastic is already the frugal choice over fiberglass or balsa turnings for this diameter.
Fin stock — birch plywood, 3.2 millimeter (1/8 inch) thickness	Apogee Rockets, Wildman Rocketry, or a local hobby/craft store	$6–12 per sheet (enough for several fin sets)	A local hobby/craft store's birch plywood sheet (sold for scroll-saw or model work) is often identical stock to what rocketry vendors resell, at a lower price.
Launch lugs (2×)	Apogee Rockets	$1–3 for a multi-pack	Buy the multi-pack even though you only need two — spares are pennies and useful for the next build.
Parachute — ripstop nylon	Apogee Rockets or Wildman Rocketry (pre-made), or DIY	$10–20 pre-made; $5–8 DIY from a fabric remnant	A pre-made parachute is the safer choice for a first launch since sizing/venting is already validated; DIY is the frugal option once you've flown once and are comfortable with the sewing/sizing math.
Shock cord + attachment hardware	Local hardware store (tubular nylon webbing) or Apogee Rockets	$3–6	Tubular nylon webbing from a hardware or outdoor-gear store is the same material rocketry vendors resell, usually cheaper by the foot.
Epoxy / adhesives	Local hardware store	$8–12	A general-purpose 30-minute epoxy from a hardware store performs the same as rocketry-branded epoxy for this motor class.

Subsystem subtotal (frugal): roughly $50–80, close to the original $20–40 estimate once recovery gear (parachute, shock cord, epoxy) is folded in — the original project-plan figure covered tube/fins/nose cone only.

4. Motor

Part	Cheapest Source	Frugal Price	Buying Notes
AeroTech G74 Economax (single-use, standard delay + ejection)	Local hobby shop first if one stocks rocketry motors; otherwise Apogee Rockets or Wildman Rocketry	~$52 for a 2-pack (~$26 per motor)	Motors ship HazMat, which typically adds a flat $25–30+ surcharge per online order regardless of order size. A local hobby shop that stocks AeroTech/Estes motors avoids that fee entirely — worth a phone call before ordering online, especially if you're only buying one or two motors.

Part	Cheapest Source	Frugal Price	Buying Notes
Igniters	Often bundled with the motor; otherwise a small multi-pack from the same vendor	$1–2 each if bought separately	Buy 2–3 spares regardless of source — a dud igniter on launch day with no backup is the single most common reason for a scrubbed launch.

Ordering tip: wait until close to your launch date to buy the motor (per the project plan), but call a local hobby shop while you're still finalizing your OpenRocket sim — knowing whether local pickup is an option can change whether it's worth ordering the motor together with your next airframe-parts order to split a HazMat fee, or separately from a local shop with no fee at all.

5. General Frugal-Buying Principles

• **Consolidate AliExpress orders.** Put every commoditized electronics part (Nano, BMP388, MPU6050, GPS, SD module, LoRa pair, regulators) into one cart from one or two sellers where possible. Shipping cost and wait time are largely fixed per package, not per item, so five items in one order is far cheaper per-part than five separate orders.

• **Buy consumables in bulk, always.** LEDs, resistors, jumper wires, and screws cost pennies more per unit in a pack of 20–50 than buying exactly one. Since you'll need spares for bench-testing mistakes anyway, there's no frugal reason to buy a single unit of anything under $2.

• **The motor is the one place shipping fees can matter more than the sticker price.** A $10 price difference between two motor vendors is easily erased by a $25+ HazMat shipping surcharge. A local hobby shop, even at full retail price, can beat any online vendor once shipping is included — always check locally first for the motor specifically.

• **Spend slightly more on parts that are reused for years.** The launch controller and the electronics sled are the two build categories where paying a bit extra for a sturdier switch, a name-brand battery, or a better enclosure pays for itself many times over across future flights — unlike the motor and igniter, which are gone after one launch regardless of what you spent.

• **Re-check vendor prices before each order.** Apogee Rockets and Wildman Rocketry compete on largely the same catalog of airframe parts; whichever is cheaper changes from part to part and month to month, so a quick side-by-side check before checkout is worth the two minutes it takes.

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6. Frugal Budget Summary

Category	Original Project-Plan Estimate	Frugal-Sourcing Estimate
Flight computer electronics	$25–45	$25–40
Radio (2× LoRa + ground receiver)	$15–30	included above
Launch controller (one-time, reusable)	$15–30	$30–60
Airframe + recovery materials	$20–40	$50–80
Motor + igniter	$15–25	~$26–30 (local pickup) or ~$50–60 (shipped, HazMat fee included)
Insurance spares	$10–15	$5–10 (cheaper since most spares are folded into bulk buys above)

The launch controller and recovery-gear categories run a bit higher here than the original quick-reference table because this guide assumes a durable, multi-year reusable build and includes parachute/shock cord/epoxy that the original table's airframe line didn't itemize separately. Total first-build cost lands in a similar overall range once local motor pickup is used — the real savings from this guide come from *where* each dollar goes (bulk electronics, local motor pickup) rather than a lower grand total.

Generated as a frugal-sourcing companion to the project's electronics viability research and motor/electronics sourcing reference, for the 3,500 ft no-certification rocket build.